JAVASCRIPT | ASSIGNMENT 5

Function Architecture and Scope

Task: Define a standard function and an arrow function that accept parameters (e.g., length and width) with default values to calculate the area of a rectangle. Demonstrate function and block scope limits.

CODE

```
// Assignment 5

// Standard function with default parameters
function rectangleArea(length = 5, width = 4) {
  return length * width;
}

// Arrow function with default parameters
const arrowArea = (length = 5, width = 4) => length * width;

console.log("Standard Function Area (Default 5x4): " + rectangleArea());
console.log("Arrow Function Area (Custom 10x3): " + arrowArea(10, 3));

// Scope demonstration
function scopeTest() {
  let functionVar = "I exist inside the function";
  if (true) {
    let blockVar = "I exist only inside this block";
    console.log(blockVar); // works here
  }
  console.log(functionVar); // works here
  console.log(blockVar);    // ReferenceError
}

try {
  scopeTest();
} catch (error) {
  console.log("Scope Test Error: " + error.name + ": " + error.message);
}
```

OUTPUT

```
Standard Function Area (Default 5x4): 20
Arrow Function Area (Custom 10x3): 30
I exist only inside this block
I exist inside the function
Scope Test Error: ReferenceError: blockVar is not defined
```

NOTES

- Default parameters apply only when an argument is missing, so rectangleArea() uses 5 and 4.

- let and const are block-scoped, so blockVar does not exist outside the if block.
- try...catch prints the error cleanly instead of crashing the script.